

# An Overview of Wireless Body Area Networks for Mobile Health Applications

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## ABSTRACT

Wireless body area networks (WBANs) are an emerging technology expected to revolutionize mobile healthcare via the real-time monitoring and analysis of medical data. This article describes the generalized WBAN concept and examples of WBAN medical applications for a variety of users and scenarios. The properties of wireless communication channels used in WBANs — specifically the human body communications channel and the 60 GHz Millimeter-wave (mmWave) — highlight the challenges facing modern WBAN development. Several research topics related to WBAN implementation in 5G networks are examined, including simulating extremely high-frequency communications, routing protocols, energy harvesting, tailoring WBANs to individual users, and user satisfaction. Future WBAN developments related to 6G networks and quantum technologies are also explored to demonstrate the potential future of mobile health.

## BACKGROUND

Wireless body area networks (WBANs) are short-range, low-power, reliable sensor networks designed and widely used for health monitoring services. Various sensor devices near, on, or within the human body record a user's vitals (e.g., blood pressure) or information from the surrounding environment (e.g., cochlear implants). These sensors then stream data wirelessly to a designated coordinator device that connects to the Internet, where it can be received by remote servers. Real-time monitoring and predictive modeling services run by these servers can alert the user or medical professional upon making a diagnosis (e.g., early signs of cancer) or detecting an emergency (e.g., warning signs of a heart attack).

For example, a WBAN user struggling with a cocaine addiction may experience an increase in locomotion and electrodermal activity (EDA) while experiencing a decrease in skin temperature when using the drug [1]. Upon detecting the appropriate changes to the user's locomotion, EDA, and skin temperature from the corresponding data streams, the remote server could immediately notify medical professionals of the user's cocaine use. This can be extended to anticipating usage before it occurs through the recognition of various biophysiological patterns indicative of cravings, enabling pre-emptive treatment or intervention. The WBAN can also circumvent the

user's failure to accurately self-report symptoms or substance use to medical professionals, such as misremembering craving symptoms or denying recent cocaine usage.

Medical WBAN applications are expected to be extremely useful for a variety of users, particularly vulnerable populations such as the elderly, newborns, and the chronically ill, and professionals in high-risk situations (e.g., soldiers, firefighters, police officers). User-directed services are also possible, such as assisting users with disabilities (e.g., navigation systems for visually impaired users) or determining causes of unhealthy habits (e.g., recognizing daytime activities that lead to a poor sleep schedule). Additionally, these applications are not limited to users who may be considered "at-risk" — users may not recognize an illness if they appear asymptomatic or cannot recognize early symptoms (e.g., contagious diseases such as COVID-19), may desire monitoring for hazards and unforeseeable accidents (e.g., car collisions in low-traffic areas), or may wish to monitor their vitals, habits, and history in the day-to-day (e.g., athletes in training). Finally, non-medical applications for large data services such as voice or video streaming may also be implemented for information, education, and entertainment purposes.

There are several requirements for the effectiveness of an individual WBAN [2, 3]:

- WBANs must not harm nor inconvenience users. Health risks from the radio frequency of continuous wireless communications, heat generated from devices, possible obstruction by implanted devices, and methods needed to retrieve implanted devices such as surgery must be considered. In addition, WBANs should feel unobtrusive — ideally, devices will be "invisible" to users.
- All devices and batteries must be rechargeable and easy to replace. This is typical in mobile devices, but critical for a constant monitoring service where any downtime may lead to faulty or missed diagnoses. Accessing devices may also be an invasive, difficult, or dangerous process, particularly for implants.
- All services must be secure. Wireless communications are particularly vulnerable to outside attacks that can compromise data or deny service, as well as interference from other devices and environmental noise. Remote servers must be secure as they contain sensitive personal data and are responsible for interpretation and notification services.

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- Each WBAN must be tailored to its user. Sensor usage, position, and technical specifications may vary depending on each user's needs, habits, abilities, body measurements, etc., as will services provided by remote servers. Coordinator devices may also be dedicated devices or a user's personal IoT device, most likely a mobile phone.

Depending on its function, sensor devices may be worn externally, attached to or implanted within the surface of the user's body; implanted internally through surgery; or administered orally. As vitals change at distinct frequencies across different value ranges, the data rate for different kinds of sensor devices varies as well (Table 1). Power consumption also increases with higher sensor frequency and larger value ranges due to greater packet totals and packet sizes, respectively.

As of 2020, WBANs are still in the experimental phase of development. The gradual deployment of 5G cellular networks is encouraging, given the need for high-volume data communications. Exploring the intricacies of extremely high-frequency bands, generally around 60 GHz, is an essential part of ongoing research, as is optimizing power consumption through techniques such as energy harvesting and routing algorithms.

## WBAN DEVICES AND COMMUNICATIONS

As defined in the IEEE 802.15.6 standard [4], WBAN communications can be described under seven scenarios based on the location of each device and the communication medium between them. We generalize both devices and communications as belonging to the internal and external domains based on location.

### INTERNAL DEVICES AND COMMUNICATIONS

Internal devices offer the possibility of examining or interacting with human organs and systems that may be inaccessible externally (Fig. 1). It is essential, however, that the placement and potential movement range of an internal device is carefully considered given the vulnerability of internal organs and the potential difficulty in extraction. The devices themselves must also be durable enough to withstand the heat, pressure, and chemical composition of its potential environment such that it neither fails to operate nor pollutes the body with material shedding. It is expected for implanted internal devices such as pacemakers to have lifetimes of more than 5 years; by comparison, accessible external devices such as electrocardiograms may be expected to operate in the realm of several days or weeks [5].

The internal communication channel is formally recognized as the Human Body Communications (HBC) Physical (PHY) layer, where the body itself is used as a transmission medium. First proposed by Zimmerman in 1996 [6], modeling HBC remains an ongoing focus of WBAN research [7] [8]. In order to safely and reliably penetrate dense body materials such as muscle, fat, and organ tissue, HBC PHY is generally limited to frequency bands between 5 and 100 MHz with bandwidths of about 3 to 5 MHz. Path loss for HBC PHY is difficult to approximate for several reasons:

- Sensor nodes can have a variety of positions on, around, or within the human body based on what characteristics they monitor (i.e.,

| Application           | Data Rate   | Frequency     | Latency | Accuracy   | Energy   |
|-----------------------|-------------|---------------|---------|------------|----------|
| Brain depth simulator | <16 kbps    | 130 Hz        | <150 ms | —          | —        |
| Cochlear implant      | <1Mbps      | 5, 12, 49 MHz | <150 ms | —          | —        |
| Voice                 | 100 kbps    | —             | <250 ms | —          | —        |
| Motion sensor         | 4.8–35 kbps | 30–100 Hz     | <250 ms | 12–16 bits | —        |
| Video streaming       | <10 Mbps    | —             | <250 ms | —          | High     |
| Endoscope capsule     | >2 Mbps     | —             | <150 ms | —          | High     |
| Blood pressure        | <10 kbps    | <100 Hz       | <150 ms | 12 bits    | High     |
| ECG                   | 3 kbps      | <500 Hz       | <150 ms | 12 bits    | High     |
| Audio streaming       | 1 Mbps      | —             | <250 ms | —          | High     |
| Pacemaker             | <1 kbps     | <500 Hz       | <150 ms | 12 bits    | Low      |
| Drug delivery capsule | <320 kbps   | —             | <150 ms | —          | Low      |
| Blood flow rate       | 480 kbps    | <40 Hz        | <150 ms | 12 bits    | Low      |
| Temperature           | 120 bps     | <1Hz          | <150 ms | 12 bits    | Low      |
| Glucose level sensor  | <1 kbps     | <50 Hz        | <150 ms | 16 bits    | Very low |

From Olatinwo et al. [2], ordered by energy consumption.

TABLE 1. WBAN applications and communication properties.

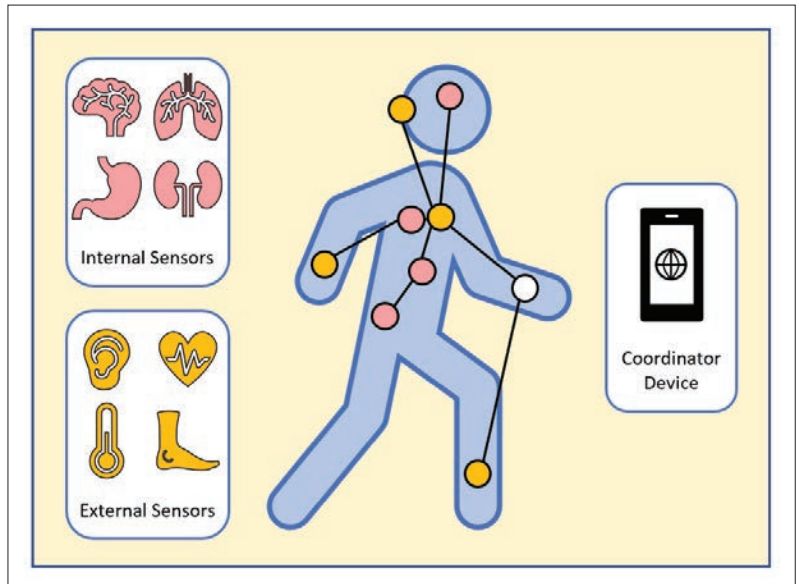


FIGURE 1. Diagram of the general WBAN structure at the user level.

temperature and heart rate).

- No two bodies are the same. Even individuals of roughly similar size and shape may have a difference in resonating frequency or conductivity based on muscle, fat, bone, organs, other implants, and so on.
- An individual's posture, clothing, accessories, and surroundings can change frequently and unpredictably; all of these factors can affect signal propagation.

An average path loss of about 75 dB with an expected fluctuation range of about 68 to 82 dB has been estimated by Hwang, *et al.* [7]. As a consequence of the limited bandwidth and unpredictable path loss, the data rates for HBC PHY have a low ceiling. The Frequency Selective Digital Transmission (FSDT) scheme under the IEEE 802.15.6 HBC PHY standard states a maximum of 1.3125 Mb/s, though sensors designed for short-term use and easy retrieval such as endoscope capsules — which have data rates over 2 Mb/s — may exceed this amount.

### EXTERNAL DEVICES AND COMMUNICATIONS

External devices include any wearable device (e.g., wristbands or devices held in pockets or bags) and devices attached externally to the body surface, including a user's coordinator device. While these devices are more easily accessible and generally in less vulnerable locations, they must still

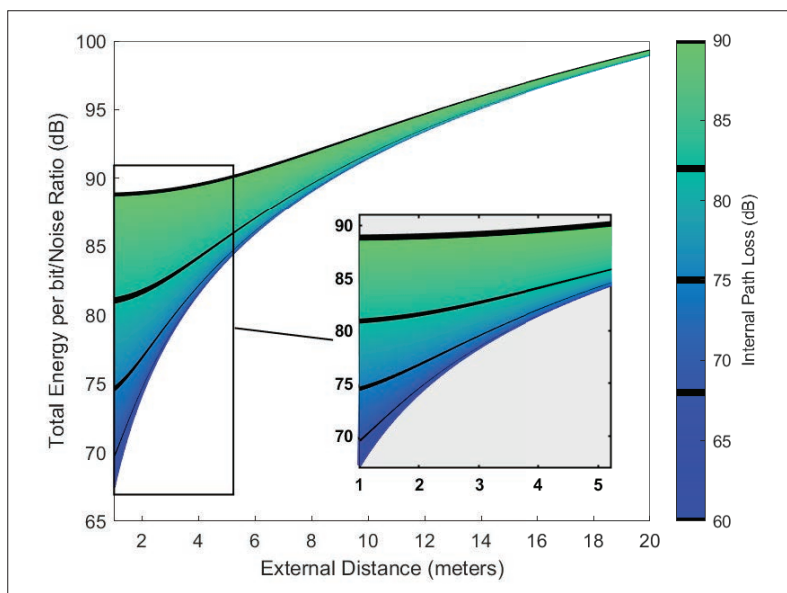


FIGURE 2. Relationship of energy costs with internal and external path loss in non-homogeneous end-to-end-relay environments.

consider the user's safety and convenience. In addition to various sensors and the coordinator device, relay nodes may also be included to mitigate power consumption and improve stability between distant devices.

Traditionally, the 2.4 GHz microwave Industrial, Scientific, and Medical (ISM) band is the standard band for WBANs [4]. However, the growing requirements for the real-time high-speed communication of WBANs prove that the 2.4 GHz microwave band is insufficient for the PHY channel of future WBANs. In addition, the 2.4 GHz band is used by many existing services that would interfere with the reliability of WBAN communications.

The 60 GHz mmWave band is a promising alternative, offering high data rates for low power costs in short-range networks [9]. This comes with the drawback of limiting the effective transmission range, as higher frequency communications are more susceptible to propagation limitations such as environmental noise, reflection, and diffraction. Communications within the range of the human body will usually be within 1 meter — well within the reliable range of about 5 meters for the 60 GHz mmWave — but communications between a coordinator device and a nearby wireless router may be as high as 20 meters in non-line-of-sight (NLOS) environments. Interference from other nearby devices (e.g., other WBAN users) could also impair communications if the allocated 60 GHz mmWave band becomes overcrowded like the 2.4 GHz band. Fortunately, methods used in 5G networks, such as beamforming — where the signal is focused toward its destination rather than sent in all directions — and massive multiple input/multiple output (MIMO), where devices have tens or hundreds of transmitting and receiving antennas instead of two or four, can offset these issues by improving transmission speed and accuracy [10].

#### NON-HOMOGENEOUS END-TO-END RELAY ENVIRONMENTS

A non-homogeneous end-to-end relay environment is any end-to-end communication involving both an internal channel and an external chan-

nel [11]. For any device communicating along an internal channel (including all internal devices), the end-to-end relay environment is inevitably non-homogeneous. Communications between internal and on-body external devices utilize the HBC PHY channel used in internal communications [8]; internal to off-body external communication is not recommended. The end-to-end data rate (i.e., the “total” data rate) under any relay environment is determined by the data rate and transmission time for each link. Since internal devices are typically expected to have low data rates, internal link restrictions may inhibit improvements made to external links. Optimizing the trade-off between data rates and power consumption for these environments is essential.

A modification of a model from Liu *et al.*, [11] was used to simulate a simple two-link non-homogeneous end-to-end relay environment for a typical WBAN user to estimate the total energy per bit to noise ratio for the entire system. Analysis of this model suggested that the path loss from either link will be the greatest contributor in power consumption when the data rate is within an established limit appropriate to its bandwidth — approximately 1.3125 Mb/s for the HBC internal channel and 10 Mb/s for the 60 GHz mmWave external line-of-sight (LoS) channel. The HBC internal channel in this model assumes internal path loss values between 68 and 82 dB, according to Hwang *et al.*, [7] reflecting the uncertainty of each user's physical composition and posture. Since the environment of the external channel is comparatively simple, the external path loss was determined from the distance between the coordinator device and a wireless router within a range of 1 to 20 meters.

In Fig. 2, a nearly 30 dB increase in the energy per bit to noise ratio between the expected best- and worst-case scenarios can be observed, where the external link's path loss (as determined from distance) carries more weight in most situations. Notably, this simulation is representative of a somewhat ideal environment (indoor LOS) — most other environments will experience greater path loss at similar distances, effectively shortening the range of efficient transmission and further increasing the importance of the development of external channel communications.

#### MODERN DEVELOPMENTS

As part of the development of WBAN devices and communications, modern research developments are primarily focused on quality of service (QoS) factors, such as improving network stability, optimizing power consumption, evaluating user satisfaction, and refining predictive modeling methods. The topics of research are generally focused on the following areas: exploring the characteristics of the 60 GHz mmWave and other frequencies on the extremely high-frequency (EHF) band used in 5G networks; creating and refining routing protocols to reduce power consumption from signal transmission; developing energy harvesting techniques to extend battery life; tailoring WBAN effectiveness and safety to individual users; running field experiments with human participants to gauge user satisfaction with WBAN devices; and determining the accuracy of predictive models.

As mentioned previously, the 60 GHz mmWave band is expected to fulfill the high data rate and low power requirements for external communications in medical WBAN applications. However, the true capabilities and limitations of this band have yet to be fully explored. Development of WBAN communications under 5G networks relies on large-scale testing of 60 GHz communication testing environments. As with any frequency band, this includes accessible methods for simulating and emulating software and devices.

In Bruendl *et al.*, [9], the potential of the Universal Software Radio Peripheral (USRP) from Ettus Research as a hardware emulator used with free software development environments (e.g., GNU Radio and MATLAB) is explored as an affordable means of emulating multi-node WBANs at different frequencies. Note that no publicly released node networks operated at 60 GHz at the time of the experiment's publication; most wireless medical devices operated under the 2.4 GHz band with little consideration toward overpopulation or improvements from the imminent arrival of 5G networks. This emphasizes the role simulations play in the development of disruptive technology and underscores the importance of its accessibility.

A continuation of this research led to the development of an emulation platform that evaluates the real-time transmission of WBAN data streams under various frequencies near 2.4 GHz [12]. The pipeline for this platform uses real sensor node data streams (e.g., skin temperature, locomotion, EDA change) from Carreiro *et al.*, [1] in its replication of data collection. Data is sent every 0.5 seconds using the USRP with GNU Radio as smaller chunks of data with additional padding to resolve file size limitation issues. Each chunk is checked for errors and resent if a sufficiently high error rate is observed. Upon validating every chunk, the data set is reconstructed and run through a real-time peak detection algorithm to detect cocaine usage. They conclude that the bit error rate of transmission is reduced when communicating over a higher frequency, indicating the potential of the 60 GHz mmWave band.

### ROUTING PROTOCOLS

There exists a trade-off between end-to-end transmission speed and stability with the power consumption and subsequent heat generation of devices. Faster, more reliable communications have greater energy costs, especially as the distance between nodes increases, when communicating in a lossy environment, or when the directivity between antennas is low. Routing protocols attempt to optimize the end-to-end transmission path (from sensor to coordinator device) by determining the most efficient communication path available at the expense of computing time and power. Qu *et al.*, [13] outline and compare various methods classified into several generalized methods:

- *Posture-based routing* determines the most efficient path from the network topology of the human body in different posture positions. Based on the regularity of various body movements, a node could predict the positions of other nodes in advance, simplifying

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ing the dynamic environment of the human body and enabling beamforming techniques.

- *Temperature-based routing* selects paths based on the temperatures of other nodes in order to limit overheating. Low temperature nodes are prioritized, allowing high temperature nodes to cool down.
- *Cross-layer routing* collaborates between the different communication layers (e.g., MAC, Network) to select a path based on the data itself. This allows for prioritization and customized services for different data sources.
- *Cluster-based routing* divides large networks into clusters based on proximity. Communications are limited to nodes within a cluster, except for a designated cluster head that handles inter-cluster communications.
- *QoS-based routing* considers the data priority, energy efficiency, link reliability, and data transmission reliability, low transmission delay, node temperature, data security, and other QoS factors when selecting a path.

Other unclassified routing methods include multipath-based routing, which sends packets of data through different transmission paths to balance energy consumption across nodes and increase stability, and mobile sink-based routing, which intermittently assigns sink nodes to ease the strain on hotspots and distant nodes.

### ENERGY HARVESTING

In addition to reducing power costs, the battery life of a WBAN device can be extended by harvesting energy from the body and its environment [14]. The availability of a given energy source can vary depending on a user's body shape, habits, and environment, or the size, location, or function of a device:

- *Solar or light energy* can be obtained by collecting sunlight through small photovoltaic (PV) cells on external devices. This offers a clean, low-cost solution that can generate up to 100 milliwatts per cm<sup>2</sup> in direct sunlight or about 100 microwatts per cm<sup>2</sup> indoors during daylight hours, giving it the highest potential power density of all sources.
- *Vibration or mechanical energy* from the user's movement (including from locomotion and physiological processes like breathing) can be collected by devices equipped with a piezoelectric transducer. The unpredictability of a user's habits — consciously or otherwise — limit the availability of this source depending on a device's location, though it offers the highest potential voltage output among all sources at about 20 V.
- *Thermal or heat energy* is an attractive source given the human body's ability to regulate heat. Using a thermoelectric generator (TEG), the difference in temperature between a user's skin and their environment can be made into a consistent electrical source, where a larger difference correlates to a larger output voltage.
- *Radio frequency (RF) energy* from ambient radio signals is a promising source given the

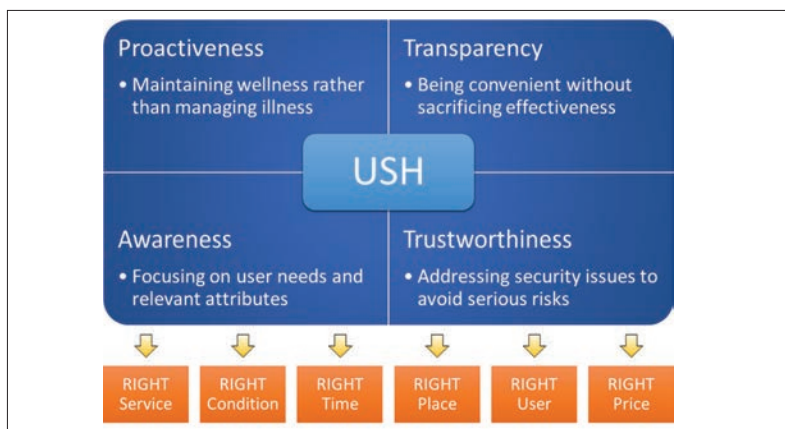


FIGURE 3. The Ubiquitous Sensing for Healthcare (USH) framework goals (top) and healthcare supply chain integrity values (bottom) as described by Zhang *et al.*, [3].

increasing popularity of wireless devices and the indifference to natural environmental conditions such as weather. While the output voltage from this source is low, the ease of implementation and high compatibility with WBAN devices have made it a popular subject of recent research.

Demir *et al.*, [14] discuss the implementation and effectiveness of energy harvesting methods for these sources at large and WBAN devices specifically. Hybrid energy scavenging methods are also explored as a means of maintaining a consistent output voltage and thus, improving the reliability of WBANs in different scenarios (e.g., a pulse monitor that relies on its TEG in an air-conditioned office and its solar cell at a public park). It is expected that the hybridization of these individual methods will become the focus of future WBAN energy harvesting methods.

### TAILORING WBANS TO INDIVIDUAL USERS

Personal devices depend heavily on the nature of an individual user [3]. The effectiveness of a WBAN is particularly dependent on a user's body, habits, and needs such that no two WBANs will be the same. Generally, WBAN research has focused on improving device and network performance such that successful WBAN implementation should be possible under any user environment. However, many of these improvements are simulated under relatively simple scenarios (e.g., an adult male of average height and weight standing in an empty room). Representing real users is a necessity for accommodating an infinitely varied user base.

General limitations can be inferred from a user's physical attributes and application needs. A child will likely require fewer relay nodes than an adult given the shorter distance between sensor locations. Prosthetic limbs, aside from the point of contact with the user, would only be compatible with locomotion and environmental sensors. However, they could also tolerate higher device temperatures and wireless communication frequencies than natural limbs. Users who require audio and video streaming applications depend on higher data rates and energy costs than users who only wish to monitor vitals such as temperature and glucose levels. Generalized models for managing these factors are plausible given their

fixed nature, particularly with developments in HBC modeling [8].

Anticipating an individual user's habits is a primary challenge in customizing a WBAN [3]. Changes in posture directly affect the distance, antenna directivity, and path loss between nodes, impacting the energy required to reliably communicate, and subsequently, the device temperature. Changes in the user's environment may impact the path loss due to noise (or the lack thereof) or the availability of wireless networks. Predicting user habits form the basis of several routing protocols and energy harvesting methods [14]. Accurate habit prediction relies not only on effective algorithms, but on the user behaving normally while wearing a WBAN.

### USER SATISFACTION

Ideally, WBANs could be as prevalent as smart-phones, assuming user expectations for efficacy, safety, convenience, and affordability can be met. Zhang *et al.*, describe the Ubiquitous Sensing for Healthcare (USH) framework as four goals for understanding human values in WSN-related healthcare services: proactiveness, transparency, awareness, and trustworthiness [3]. Achieving these goals should ensure that the service, as a collection of products, can meet the healthcare supply chain integrity values seen in Fig. 3.

5G communications and cloud computing services are generally well-suited to meeting these goals in WBANs, though achieving transparency stands out as a unique challenge. As mentioned previously, the ideal WBAN is "invisible" to the user so that the device does not affect their normal behavior. Improving performance may increase visibility in some cases, such as when adding more sensor nodes or increasing power consumption. Individual users may also object to devices in certain locations — implant usage, in particular, is often a sensitive issue. Solutions typically focus on the development of wearable devices given the high potential for user convenience.

**Evaluating User Satisfaction:** Carreiro *et al.*, explore the effectiveness and transparency of a mobile biosensor system to detect cocaine use [1]. In this experiment, 15 consenting participants with an existing diagnosis of cocaine dependency were selected to wear a portable biosensor strapped to their non-dominant wrist. The devices measured EDA, skin temperature, and 3D locomotion (five attributes total) 20 times per second during the 45-day trial period. Participants were expected to meet with study personnel twice a week to report events over the previous three- to four-day period and to provide a urine sample for drug screening. Unlike WBANs, participants needed to remove and recharge the device while sleeping and only wear the device while awake. Sensor data was also stored locally for analyses conducted after device retrieval rather than in real-time. With regards to transparency, the results were generally positive:

- 1 out of 15 found the sensor annoying to wear.
- 1 out of 15 had problems using the device.
- 1 out of 15 had clothing items that made wearing the sensor uncomfortable.
- 14 out of 15 would consider wearing a second sensor on the other wrist.
- 15 out of 15 would be willing to wear the sensor for a longer period.

The presence of the device was noted to inhibit drug use in seven out of 15 individuals “as it made them more mindful of their actions and ‘would look at the sensor and would think about prison.’” [1] Participants also reported that friends and family were mostly either curious or indifferent about why the participant was wearing the biosensor. Overall, most participants did not report behaving differently — it was “life as usual.”

**Developing Flexible Devices:** Adding “hard” devices worn as accessories should be a consistently convenient solution for including sensors at various locations (e.g., necklaces for monitoring the upper chest). However, many users may not prefer to wear a particular accessory or may not wish to wear too many accessories. They may also benefit from sensors in locations where a hard material may be uncomfortable, such as the abdomen. For these reasons, the development of flexible sensors and antennas for use with clothing-based devices has been a popular subject of research.

Alhawari *et al.*, [15] presents a low-cost meta-material-based flexible antenna made of denim and felt substrates usable in WBANs. The proposed antenna is a mere 10 x 15 mm<sup>2</sup> in size and 2.2 mm thick and can be attached to any garment. The operating bandwidth is between 3 to 40 GHz (peak transmission coefficient at 33 GHz) with a maximum gain of 8.85 dBi. The Specific Absorption Rate (SAR) value is also below the recommended 2 W/kg in this range for the standard weights of 1g and 10g, ensuring no harm to a user’s skin, breast, fat, or muscle tissue. Simulations with multiple antenna arrays (sizes of 4, 8, and 12 antennas) placed near the surface of a human breast (150 mm diameter) were also able to detect tumors as small as 4 mm in diameter, with larger array sizes improving the clarity of reconstructed images.

## FUTURE DEVELOPMENTS AND 6G NETWORKS

5G communications are considered an essential foundation for the realization of WBANs [10]. However, sophisticated applications that require extremely high data rates, low latency, and ultra-reliable communications may be possible under the sixth generation (6G) of wireless networks. Considerations for 6G systems are presently underway to support the exponential growth of mobile systems. According to the International Telecommunication Union (ITU), global data rates are expected to rise to about 5016 exabytes (EB), about 1 million terabytes among 17.1 billion subscribers in 2030 compared to 7.462 million EB among 5.32 billion subscribers in 2010.

Current proposals for 6G networks are characterized by artificial intelligence integrated communication and convergence of devices and networks (as under IoT) with people and physical objects under an Internet of Everything (IoE). This unification allows for the realization of technologies such as advanced autonomous devices for the rapid deployment of emergency medical services, haptic and wireless brain-computer communications to assist the visually impaired, or extended reality (XR) services for remote surgeries.

6G wireless networks are expected to rely on terahertz (THz) communications to meet data rate demands exceeding the capabilities of the 60 GHz mmWave band. Potentially, they may utilize Optical Wireless Technology (OWT) for

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frequencies beyond the RF spectrum, such as the infrared, visible light, and ultraviolet spectrums. Implementation of massive MIMO technology through Intelligent Reflecting Surfaces (IRS) will be essential to support such high data rates while minimizing transmission power costs.

## QUANTUM TECHNOLOGY IN 6G NETWORKS

One of the cornerstones of 6G networks will be the realization of quantum technologies, specifically quantum computing and quantum cryptography. Quantum technology utilizes the characteristics of quantum states, where particles exist as probabilities of their possible values. It is reported that measuring a particle in a quantum state causes the particle to “collapse” into a classical state — specifically, as one of its possible values. For example, the probability of a coin toss landing on heads or tails would be treated as the quantum state, where the result might be either outcome, the act of flipping the coin would be considered the measurement, and the observed result would be considered the classical state, which must be either heads or tails. Furthermore, the particle may become “entangled,” where there exists a linear dependency between one particle and any number of others — like how drawing a red card from a full deck of cards affects the probability that the next card will be black. This allows for changes in one particle’s quantum state to predictably change the others, though the entangled state ends when a particle is measured. Entanglement can also persist regardless of distance, allowing for a process known as *quantum teleportation*, where the changes to all entangled quantum states are truly instantaneous rather than bound by the speed of light.

## QUANTUM APPLICATIONS IN WBANS

**Quantum Computing:** Quantum computing is expected to offer significant improvements to the speed and accuracy of large-scale calculations such as those expected in medical WBAN applications.

It should be noted, however, that quantum computers are not a replacement for most classical computers, particularly for mobile devices used in WBANs. The need for extremely stable environments typically induced by large, expensive dilution refrigerators limits their accessibility. But in coordination with massive data streams sent from WBAN users, remote quantum computers would enable machine learning and artificial intelligence methods to produce real-time results of various analyses. These results should also be more accurate than those provided by classical algorithms dependent on lossy compression methods to minimize computation time (e.g., averages of ECG measurements over longer intervals).

**Quantum Cryptography:** Quantum cryptography can assure the security of medical WBAN applications by utilizing the inherently unpredictable nature of quantum states in the encryption process. Thanks to the no-cloning theorem, which states that an arbitrary unknown quantum system cannot be copied or amplified without disturbing its original

state, stealing quantum information is as difficult as stealing the quantum device itself. Secure communication between a pre-designated number of users is also possible through entanglement.

An alternative method for WBANs to utilize quantum cryptography is quantum random number generation (QRNG) for the generation of unpredictable classic cryptographic keys. This utilizes low-power, small-size chips that generate truly random number strings using any light source and a CMOS image sensor. Unlike the previously discussed quantum technologies, QRNG is highly compatible with WBAN user devices, particularly external devices that can easily be accessed in the event of a light source's failure. Both QKD and QRNG merely provide methods for creating unpredictable cryptographic keys used with classical data. These methods cannot be considered truly secure for WBANs as they both retain a dependency on classical wireless communications. Furthermore, the advent of quantum computers may prove to render classical encryption useless as attackers would be capable of deciphering encrypted data.

## CONCLUSION

The arrival of 5G networks in 2020 sets the stage for real-world implementation of medical WBANs. Challenges such as optimizing WBAN power consumption, improving the reliability of WBAN communications, and achieving transparency with WBAN devices are still being investigated. Major research topics dedicated to establishing modern WBAN standards for 5G communication networks include WBAN modeling, routing, and energy harvesting, with future considerations made toward 6G network development and quantum technologies. Collaboration between experts in multiple fields (e.g., medical, wireless, etc.) has been and will remain a necessity in achieving these goals.

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## REFERENCES

- [1] S. Carreiro *et al.*, "Imstrong: Deployment of a Biosensor System to Detect Cocaine Use," *J. Medical Systems*, vol. 39, no. 12, p. 186, Oct. 2015.
- [2] D. D. Olatinwo, A. Abu-Mahfouz, and G. Hancke, "A Survey on Lpwan Technologies in Wban for Remote Health-Care Monitoring," *eng. Sensors (Basel, Switzerland)*, vol. 19, no. 23, p. 5268, Nov. 2019, p. 5268.

- [3] Y. Zhang *et al.*, "Ubiquitous WSN for Healthcare: Recent Advances and Future Prospects," *IEEE IoT J.*, vol. 1, no. 4, 2014.
- [4] K. S. Kwak, S. Ullah, and N. Ullah, "An Overview of IEEE 802.15.6 Standard," *Proc. 2010 3rd Int'l. Symposium on Applied Sciences in Biomedical and Communication Technologies*, Nov. 2010.
- [5] M. M. Alam and E. Ben Hamida, "Surveying Wearable Human Assistive Technology for Life and Safety Critical Applications: Standards, Challenges and Opportunities," *eng. Sensors*, vol. 14, no. 5, May 2014.
- [6] T. G. Zimmerman, "Personal Area Networks: Nearfield Intra-body Communication," *IBM Systems J.*, vol. 35, no. 3.4, 1996.
- [7] J. H. Hwang *et al.*, "Empirical Channel Model for Human Body Communication," *IEEE Antennas and Wireless Propagation Letters*, vol. 14, 2015, pp. 694–97.
- [8] M. D. Pereira, G. A. Alvarez-Botero, and F. Rangel de Sousa, "Characterization and Modeling of the Capacitive HBC Channel," *IEEE Trans. Instrumentation and Measurement*, vol. 64, no. 10, 2015, pp. 2626–35.
- [9] S. A. Bruendl and H. Fang, "Making the Switch to 5G and 60 Ghz in Mhealth Applications Using USRP Hardware," *IEEE Internet Computing*, vol. 24, no. 2, 2020, pp. 57–64.
- [10] E. Selem, M. Fatehy, and S. M. A. El-Kader, "E-Health Applications Over 5G Networks: Challenges and State of the Art," *Proc. 2019 6th Int'l. Conf. Advanced Control Circuits and Systems (ACCS) 2019 5th Int'l. Conf. New Paradigms in Electronics Information Technology (PEIT)*, Nov. 2019, pp. 111–18.
- [11] D. Liu, Y. Geng, and K. Pahlavan, "End-to-End Power Optimization in Non-Homogenous Relay Environment for Wireless Body Area Networks (WBANs)," pp. 1–5.
- [12] S. A. Bruendl *et al.*, "A New Emulation Platform for Real-Time Machine Learning in Substance Use Data Streams," *Proc. 21st IEEE Int'l. Conf. Information Reuse and Integration (IEEE IRI 2020)*, 2020.
- [13] Y. Qu *et al.*, "A Survey of Routing Protocols in WBAN for Healthcare Applications," *Sensors*, vol. 19, no. 7, 2019.
- [14] S. M. Demir, F. Al-Turjman, and A. Muhtaroglu, "Energy Scavenging Methods for WBAN Applications: A Review," *IEEE Sensors J.*, vol. 18, no. 16, 2018, pp. 6477–88.
- [15] A. R. H. Alhawari *et al.*, "Metamaterial-Based Wearable Flexible Elliptical Uwb Antenna for WBAN and Breast Imaging Applications," *AIP Advances*, vol. 11, no. 1, 2021.

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